Skill 9

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

char previous_dir[PATH_MAX] = "";

void builtin_pwd()
{
   char current_dir[PATH_MAX];

   if (getcwd(current_dir, sizeof(current_dir)) != NULL)
      printf("%s\n", current_dir);
   else
      perror("pwd");
}

void builtin_cd(char *path)
{
   char current_dir[PATH_MAX];

   if (getcwd(current_dir, sizeof(current_dir)) == NULL)
   {
      perror("getcwd");
      return;
   }
```

```
    if (path == NULL)
    {
        printf("cd: missing argument\n");
        return;
    }

    if (strcmp(path, "-") == 0)
    {
        if (strlen(previous_dir) == 0)
        {
            printf("cd: previous directory not available\n");
            return;
        }

        path = previous_dir;
    }

    if (chdir(path) != 0)
    {
        perror("cd");
        return;
    }

    strcpy(previous_dir, current_dir);
}

void builtin_help()
```

```
{
   printf("Available built-ins:\n");
   printf("  pwd      - show current directory\n");
   printf("  cd <dir>  - change directory\n");
   printf("  cd -     - go to previous directory\n");
   printf("  help     - show help\n");
   printf("  exit     - exit shell\n");
}

void builtin_exit()
{
   printf("Exiting shell...\n");
   exit(0);
}

int main()
{
   char command[100];
   char argument[PATH_MAX];

   printf("Mini Shell\n");

   while (1)
   {
      printf("myshell> ");

      if (scanf("%99s", command) != 1)
         break;
```

```
        if (strcmp(command, "pwd") == 0)
        {
            builtin_pwd();
        }
        else if (strcmp(command, "cd") == 0)
        {
            if (scanf("%4095s", argument) == 1)
                builtin_cd(argument);
            else
                printf("cd: missing argument\n");
        }
        else if (strcmp(command, "help") == 0)
        {
            builtin_help();
        }
        else if (strcmp(command, "exit") == 0)
        {
            builtin_exit();
        }
        else
        {
            printf("Invalid command: %s\n", command);
        }
    }

    return 0;
}
```

```
snigdha@Snigdha:~$ cd ~/quotes_practical
snigdha@Snigdha:~/quotes_practical$ nano cd_builtin.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o cd_builtin cd_built
in.c
snigdha@Snigdha:~/quotes_practical$ ./cd_builtin
Mini Shell
myshell> pwd
/home/snigdha/quotes_practical
myshell> cd /tmp
myshell> pwd
/tmp
myshell> cd -
myshell> pwd
/home/snigdha/quotes_practical
myshell> cd /abcd
cd: No such file or directory
myshell> help
Available built-ins:
  pwd       - show current directory
  cd <dir>  - change directory
  cd -      - go to previous directory
  help      - show help
  exit      - exit shell
myshell> hello
Invalid command: hello
myshell> exit
Exiting shell...
```